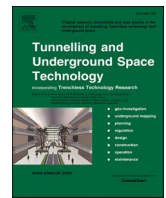




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GPRTransNet: A deep learning–based ground-penetrating radar translation network

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ABSTRACT

Ground-penetrating radar (GPR) is an essential nondestructive testing tool widely used in tunnel and road defect detection, underground object detection, and unstructured terrain perception. Currently, most GPR inversion methods based on deep learning use convolutional neural networks, which have limitations such as incomplete feature extraction and low accuracy. Inspired by advancements in the NLP field, this paper proposes a novel deep learning framework for GPR called GPRTransNet. The algorithm introduces a “wave to permittivity” translation architecture, leveraging the “memory” function of recurrent neural networks to translate GPR data into permittivity model images, similar to language translation. In addition, the inclusion of the attention mechanism significantly enhances the network’s ability to represent defects in complex scenarios, resulting in outstanding translation performance. GPRTransNet has been validated on a simulated dataset, with the newly proposed G-SSIM evaluation metric showing that the permittivity similarity of GPRTransNet-Lite reaches 98.47% and that of GPRTransNet-Pro is 99.26%. Furthermore, GPRTransNet demonstrates good translation performance on real data. Experimental results show that the results of GPRTransNet’s translation of GPR are satisfactorily matched.

1. Introduction

During highway or tunnel operation, defects such as cracks, voids, and delaminations inevitably occur because of complex factors, including geology, hydrology, and construction quality (Loupos et al., 2018; Jiang et al., 2019), seriously affecting the operational safety of highways and tunnels and potentially causing accidents (Montero et al., 2015; Feng et al., 2018). Therefore, strengthening the inspection and maintenance of highways and tunnels is particularly essential. Among the various inspection methods, ground-penetrating radar (GPR) stands out as a noninvasive detection tool (Prego et al., 2016; Alani et al., 2018) widely used in structural exploration because of its high resolution and real-time capabilities, making it a preferred tool in practical scenarios such as road, bridge, and tunnel inspections (Benedetto and Pajewski, 2013; Torbaghan et al., 2020).

With the rapid advancement of computer technology, machine learning and deep learning techniques have found extensive applications in fields such as image processing, object detection (Zou et al.,

2023), semantic segmentation (Ulku et al., 2019), and natural language processing (NLP) (Han et al., 2022; Lauriola et al., 2022). In general, the two fundamental tasks in GPR data processing are image recognition and permittivity inversion, which are used to reveal the type, location, shape, and dielectric distribution of internal targets in B-scan images. Several studies focus on the recognition and detection of GPR images. Liu (Liu et al., 2010), Dou (Dou et al., 2017), and Zhao Meng (Zhao., 2012) used traditional image processing methods such as Sobel operators, threshold segmentation, and wavelet analysis for simple target recognition in B-scan images. Ge (Ge et al., 2024) proposed a network called Wavelet-GAN to remove noise and clutter from GPR data. Zhao Di (Zhao et al., 2020) and Zhang Tianzhu (Zhang et al., 2019) used VGG16 (Simonyan et al., 2014) and AlexNet (Krizhevsky et al., 2017) networks, respectively, to classify different underground targets. Li Shufan (Li et al., 2020) improved ResNet to classify defects in concrete, achieving good test results. Besaw (Besaw et al., 2015) enhanced feature extraction and generalization capabilities by adding data augmentation to CNN-based classification (Besaw., 2016). Hu Jinpeng (Hu et al., 2006) and

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Zhou (Zhou et al., 2018) used SVM (Hearst et al., 1998) for the recognition of various GPR targets. Pham (Pham et al., 2018) and Feng Deshan (Feng et al., 2020) used Faster R-CNN (Ren et al., 2015) and YOLOv3 (Redmon et al., 2018) object detection algorithms to recognize targets in images, showing significant improvements over traditional machine vision methods. For the rebar interference problem, some scholars have adopted the strategy of eliminating rebar signals before target detection to improve the recognition effect. Wang (Wang et al., 2022) designed a GAN-based rebar clutter elimination network, RCE-GAN, to reduce the rebar interference to GPR images and improve the defect recognition accuracy. Ren (Ren et al., 2024) used an unsupervised deep learning network for B-Scan image of rebar signal suppression, using autoencoder to generate a network to remove the effect of rebar scattering, and optimizing the defect enhancement effect by feature space loss. Bao (Bao et al., 2025) proposed a global-local secondary recognition method based on improved YOLOv8 to optimize defect visibility in rebar-dense regions by performing target detection on GPR B-Scan images, and improve the recognition accuracy.

The above methods mainly focus on the task of classification or target detection in GPR B-Scan images, and their outputs are class labels or locations of defects in the B-Scan images, while it is not possible to reconstruct the image of the subsurface structure, and it is difficult to determine the shapes as well as the physical properties of the defects. In contrast, the core objective of the GPR inversion task is to estimate the spatial distribution of hidden targets based on the GPR scan data and to obtain physical parameters such as the permittivity of the target region. The inversion results can provide researchers with more detailed information about the subsurface structure, enabling them to accurately analyze the defect characteristics in the target area, which is of great value for engineering applications in road maintenance, infrastructure inspection, and traffic safety. Traditional GPR inversion methods primarily include tomography (Irving et al., 2007) and full waveform inversion (Suo et al., 2021; Lei et al., 2015), with limited progress in the field of GPR data inversion using deep neural networks. Alvarez (Alvarez et al., 2018) proposed a GPR image inversion network based on an autoencoder structure, demonstrating the effectiveness of deep learning methods in permittivity model reconstruction tasks. Liu (Liu et al., 2021) proposed a deep neural network architecture called GPRInvNet, which includes an encoder specifically designed for GPR data to achieve inversion from GPR B-scan data to permittivity images, but it did not consider the attenuation of GPR data along the tunnel depth, resulting in poor inversion results for deep surrounding rock. Ji (Ji et al., 2021) proposed the PINet network structure for GPR image inversion, and the network consists of three parts, namely, the time-dimensional data compressor, the global feature encoder, and permittivity decoder, and the inversion imaging effect of the method is improved compared to GPRInvNet, but the inversion targets are all regular geometrical diseases, and the inversion ability of the network in complex scenes has not been verified, and the real-data The inversion effect on real data needs to be improved. Ge (Ge et al., 2024) proposed a TransUNet modified with a self-attention mechanism and U-net for GPR inversion but did not verify its anti-interference capability under the influence of rebar.

Synthesizing the existing studies, it can be found that the GPR inversion task faces greater challenges compared with the traditional target detection task. The raw data captured by GPR is itself time-series information, and there is an informational relationship between the waves, while most of the existing deep learning methods are based on convolutional neural networks to invert B-Scan images, which is difficult to capture the long distance dependency information (Liu et al., 2024). Compared with convolutional neural networks, recurrent neural networks (RNN) (Zaremba et al., 2014) are more suitable for processing time-series class data (Yin et al., 2017). Meanwhile, the complexity of different scene structures also poses a challenge. The multi-layer media as well as rebar interference lead to highly complex and nonlinear waveform features, and the target defect signals are easily masked, thus making it difficult to effectively extract the disease features. In addition,

the length of the survey line is usually variable during GPR data acquisition, and the existing GPR inversion methods can only handle fixed-size data, and it is difficult to reconstruct the structural information from GPR scans of arbitrary lengths, which limits their flexibility and application scope. The main problems are described below:

- 1) Most of the existing deep learning methods use convolutional neural networks as the core, relying on local receptive fields to extract features layer by layer. As shown in Fig. 1(a), compared to RNNs, the localized nature of convolution is difficult to capture global interaction information across layers, leading to the loss of long distance dependency information, and thus learning long distance dependency is a key challenge for GPR inversion. In addition, the convolutional method is difficult to capture the propagation pattern of electromagnetic wave signals in the time dimension and is less robust to frequent inter-layer interactions and complex waveform noise, which further limits the inversion capability of the model in dealing with complex scenarios.
- 2) The GPR acquisition work usually has different survey line lengths, and the detection depth and resolution are different for different devices. Most of the existing methods can only process fixed-size B-Scan images, and cannot adapt to the size of GPR data, which lacks flexibility and adaptability. As shown in Fig. 1(b), by slicing the data and then inverting it will lead to misalignment or distortion of the merged results, which limits the adaptability of the model in different data length scenarios. In addition, the inversion method based on B-scan images ignores the original electromagnetic wave reflection characteristic information, and its input B-scan images are generated from GPR data by gray-scale mapping, which will inevitably lose some signal details and data features, especially underperforming the weak changes in low reflection intensity regions.
- 3) In a multilayer medium environment, electromagnetic wave propagation is complicated by waveform superposition, multiple reflections, and refractions, which significantly increases the difficulty of waveform feature extraction. At the same time, the high degree of nonlinear correlation between the layers of the medium exceeds the expression ability of the traditional model, and it is difficult to accurately capture the complex relationship. And the common rebar or other highly conductive objects in practical applications will have strong scattering and reflection of electromagnetic waves, as shown in Fig. 1(c), the characteristics of the diseased area are easily masked by the physical properties of the neighboring media, making the target defect signal weaker and more hidden, and the existing methods usually lack a targeted feature extraction mechanism, which makes it difficult to effectively deal with the challenges of multi-layer complex media and rebar interference.

To address the aforementioned issues, this paper proposes a novel deep learning network called GPRTransNet that is designed to translate raw GPR data from multilayered media into permittivity images. The algorithm introduces a “wave to permittivity” translation framework that translates each GPR waveform into a corresponding permittivity sequence. The network comprises an encoder-decoder structure based on gated recurrent unit (GRU), where the specially designed encoder effectively captures the relationship between abnormal GPR waveforms and adjacent waveforms. The decoder establishes a mapping between the electromagnetic signals and the target objects through the condensed “wave vector.” In addition, to enhance the model’s capability to represent complex environments, an attention mechanism (Luong et al., 2015; Vaswani et al., 2017) is incorporated to improve the focus on defect areas, thereby translating various materials and defects in the GPR data into corresponding permittivity values. This paper also introduces a novel evaluation metric specifically for GPR translation, called G-SSIM, to directly assess the translation results. GPRTransNet has been validated on simulated datasets and has good translation effect on real data. Results show that GPRTransNet matches satisfactorily for

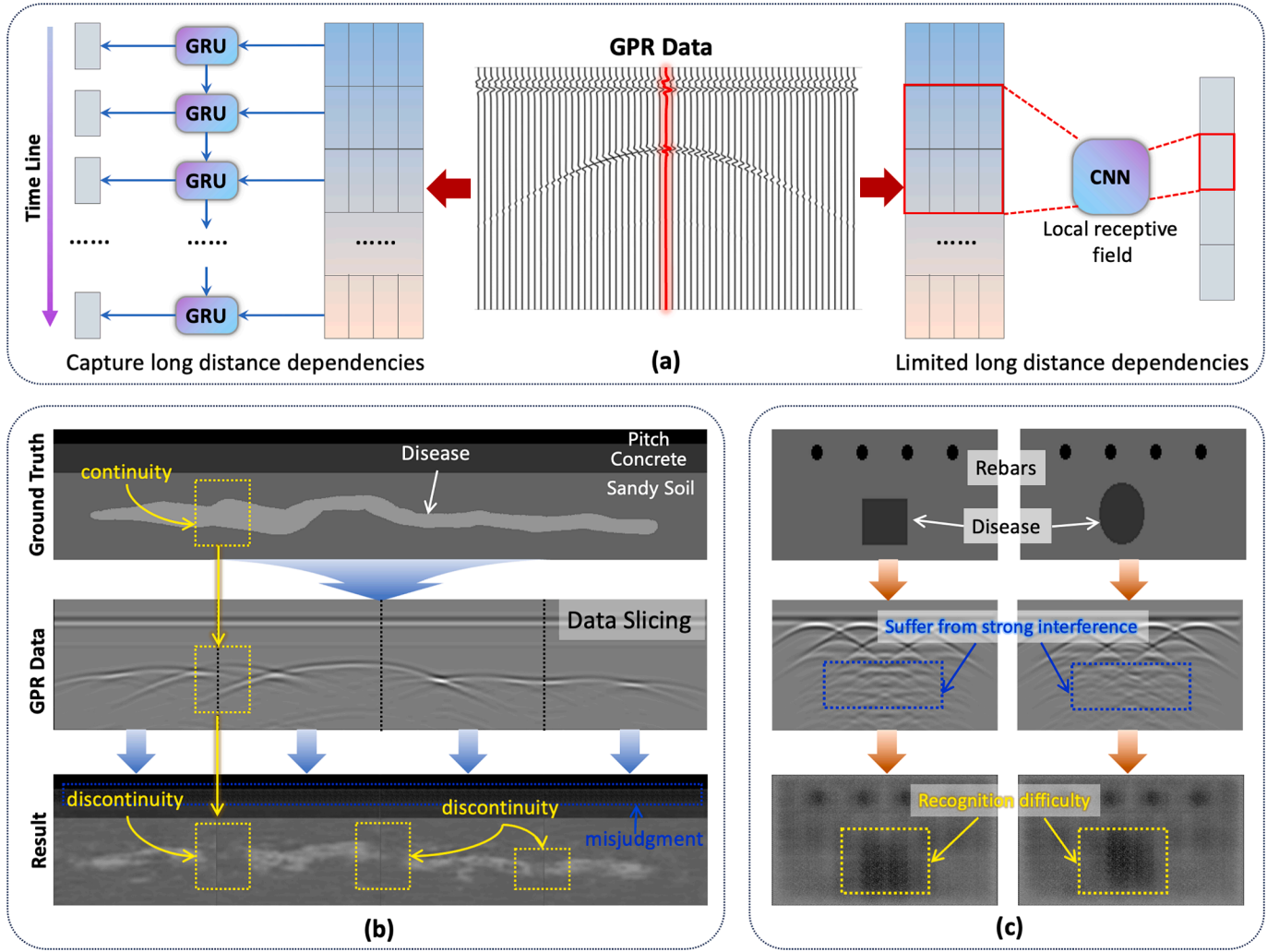


Fig. 1. Challenges in GPR inversion.

GPR translation.

The main contributions of this paper are as follows:

- 1) The proposed new deep learning framework of GPRTransNet is used to translate various types of GPR data into permittivity maps, and to the best of our knowledge, it is the first GPR inversion model based entirely on recurrent neural networks. GPRTransNet is designed to capture waveform-to-waveform correlations while using a recurrent neural network capable of passing through memory and updating mechanisms. This design allows the model to retain information about the depth direction thus solving the long distance dependency problem.
- 2) A “wave to permittivity” translation architecture is proposed, which translates each waveform of GPR into corresponding permittivity sequences, and establishes the mapping relationship between electromagnetic waves and multilayer medium parameters directly. This architecture avoids the limitation of fixed input size, and the input and output forms of GPRTransNet are always the same, which can fully adapt to the changes of trace spacing and realize the translation of GPR data of arbitrary length.
- 3) At the same time, the introduced attention mechanism enhances the network’s ability to express complex environments, and can still accurately reconstruct the internal structure of the material even in the case of strong interference such as steel reinforcement. The evaluation index G-SSIM for GPR translation is proposed, and the experiments show that GPRTransNet can translate various

permittivity models containing complex defects in different scenarios on simulated datasets, and also has satisfactory results on real datasets.

2. Translation of GPR data to permittivity maps

GPR is a technology that uses electromagnetic waves to detect subsurface medium distributions. It transmits electromagnetic waves and analyzes the characteristics of the reflected waves to describe subsurface conditions. The principle of operation of GPR involves high-frequency electromagnetic waves transmitted from the ground through an antenna into the subsurface, reflected by target objects, and then received by another antenna. The signal is processed and converted into visual data using a signal processing system. During data acquisition, the GPR moves along the survey line, collecting multiple A-scan signals that are then used to generate B-scan images that clearly display the signal characteristics of hidden objects. The principle of operation of GPR is shown in Fig. 2.

After being put into service, roads and tunnels often suffer from diseases such as deholing under the structure due to uneven settlement of the foundations and other reasons. These hollowing lesions are usually significantly different from the surrounding medium in terms of electrical parameters, and this difference is mainly manifested in the relative permittivity. Typically, the distribution of permittivity is usually represented as a spatial distribution map that directly reflects specific values at different locations. However, the way of collecting GPR data

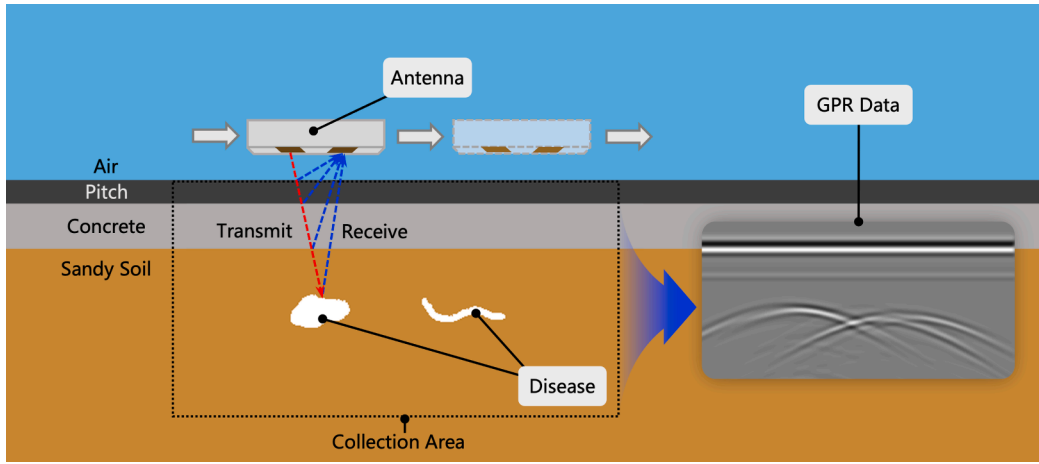


Fig. 2. Schematic diagram of the working principle of GPR.

determines its presentation as waveform signals during electromagnetic wave propagation, which record the reflection, transmission and attenuation characteristics of electromagnetic wave propagation in different media. As shown in Fig. 3, a complete waveform can be split according to the different materials that the GPR pulse electromagnetic wave signal passes through, so as to realize the mapping between the radar waveform and different material layers in the measured object, and then to extract possible voids, seepage, stratification, etc. Based on the above idea, this paper considers each GPR waveform signal as a sequence in the ‘time-depth’ dimension, and translates it point by point by deep learning network to generate the corresponding permittivity sequences. Each sequence not only reveals the permittivity sequences along the propagation path, but also reconstructs the overall spatial distribution of the target area by combining all the waveforms. This method of translating from GPR data to permittivity preserves the original characteristics of the GPR data and achieves an accurate mapping from local sequences to the overall spatial distribution.

3. Methodology

3.1. GPRTransNet components

The reflected waves collected by GPR are time-series data, where preceding and succeeding waveforms are interrelated. Compared with convolutional neural networks, RNNs are more suitable for handling time-series data. Therefore, this paper proposes a GPR translation

network called ground-penetrating radar translation net (GPRTransNet). By leveraging the powerful memory capabilities of RNNs, GPRTransNet can effectively retain temporal and depth information between GPR waveforms and layers (both laterally and vertically).

As shown in Fig. 4, GPRTransNet consists of a specially designed GPR data processing module, an encoder, and a decoder. The GPR data processing module is specifically designed for preprocessing GPR data and the established reference model. This module first reduces the number of time steps from the original sampling number to a standardized 512. This preprocessing ensures that data from various GPR devices with different sampling numbers can be normalized to a unified format, enabling consistent input to the model. Then, it maps the permittivity values of the reference model according to actual conditions and performs adaptive interpolation to match the shape of the GPR data. In addition, the data processing module normalizes the GPR data to speed up subsequent training. The encoder comprises a waveform embedding layer and an encoding network, and the decoder consists of a target embedding layer, a decoding network, and a fully connected layer.

In this paper, we introduce two versions of GPRTransNet: GPRTransNet-Lite and GPRTransNet-Pro. The encoding and decoding networks of GPRTransNet-Lite are one-layer unidirectional GRU networks with 128 neurons, which have fewer parameters and faster processing speed, enabling rapid translation of GPR data. GPRTransNet-Pro features a two-layer bidirectional GRU network with 128 neurons as its encoding network and a two-layer unidirectional GRU network with 128

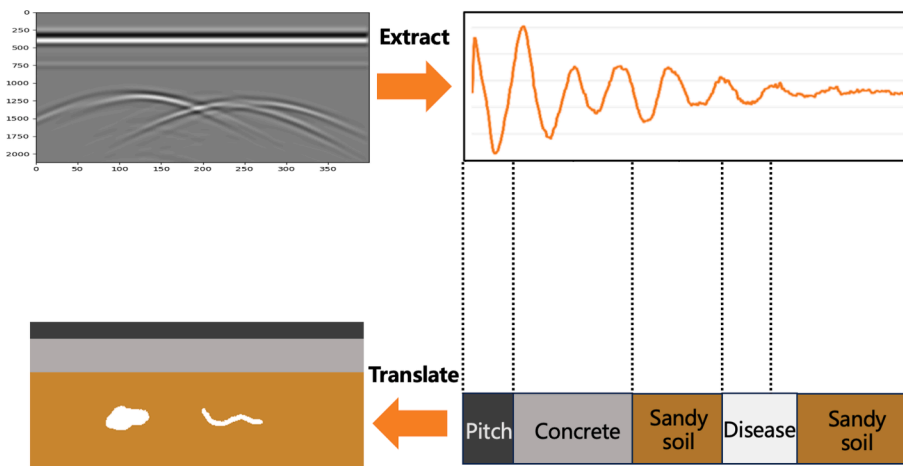


Fig. 3. GPR data to permittivity maps.

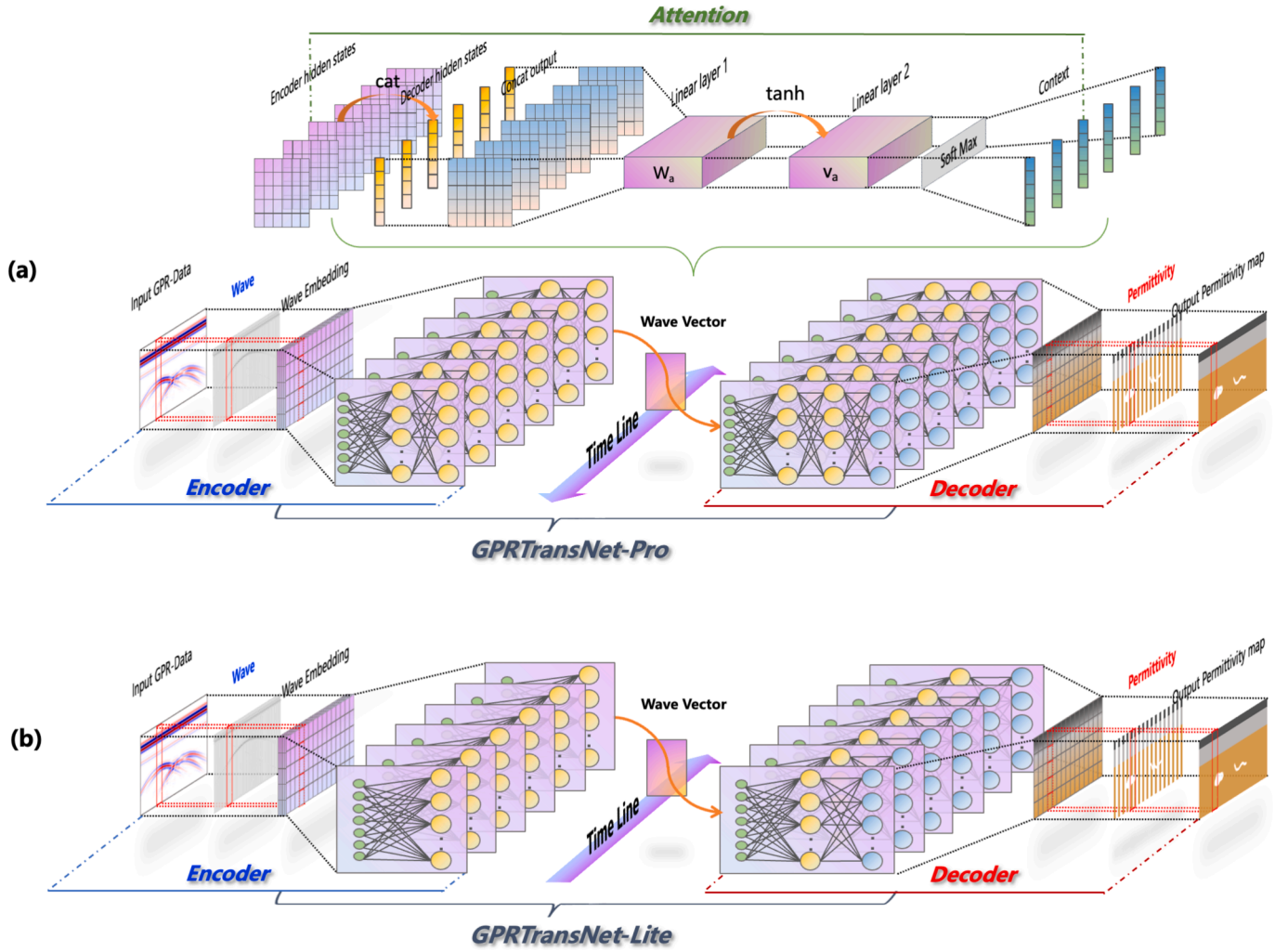


Fig. 4. GPRTransNet Structure.

neurons as its decoding network. The multilayer bidirectional GRU network introduces information flow in both forward and backward directions, processing radar sequence data in both orders. This enhances the overall semantic understanding of the sequence data, albeit at the cost of increased processing time, resulting in better translation performance.

For comparison, as shown in Fig. 5, we have also designed two convolution-based GPR translation networks, namely, GPRTransNet-Conv (GTN-Conv) and GPRTransNet-VGG (GTN-VGG). The encoder of

GTN-Conv consists of a two-layer 1-D convolutional network specifically designed to directly process each GPR waveform. The encoder of GTN-VGG is based on an improved VGG-16 network, which adjusts the structure of the input GPR data and performs effective feature extraction. The decoding networks of both convolution-based models consist of two-layer GRU networks that translate the features extracted by the encoding modules into permittivity values.

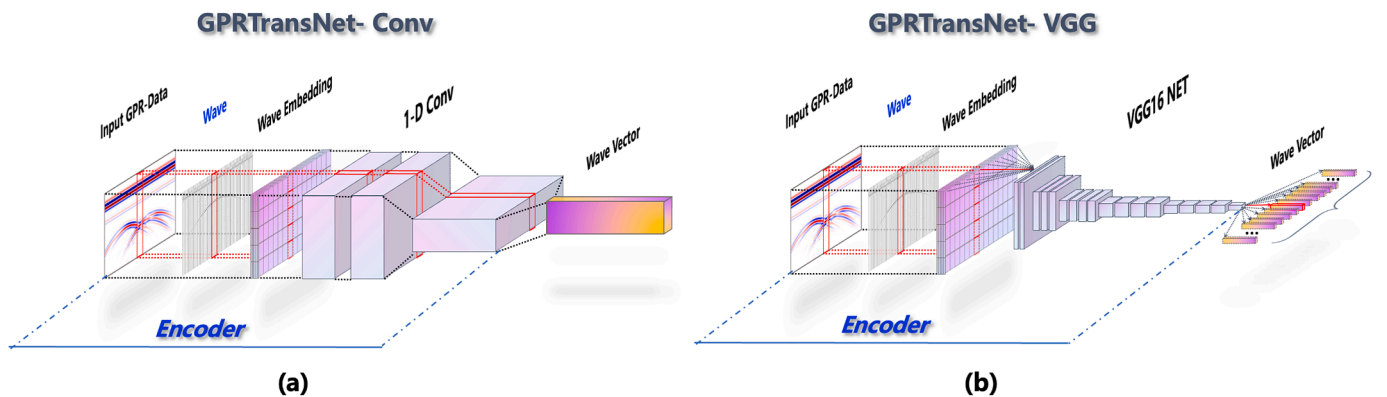


Fig. 5. Convolution-based GPRtransnet.

3.2. Wave to permittivity

GPRTNet introduces an architecture called “wave to permittivity,” inspired by the “Sequence to Sequence” architecture from the machine translation field (Sutskever et al., 2014; Cho et al., 2014). “Wave to permittivity” essentially translates a sequence of waveform features into a sequence of permittivity values. In machine translation, sequences are contextually linked, much like the continuity between sentences. It is crucial to ensure accuracy not only in the translation of individual words but also in the coherence between sentences. RNNs enable machine translation models to learn the correspondence between words and understand the semantic relationships between sentences for language translation. This concept can be effectively applied to the inversion of geological radar data.

We treat each GPR waveform as a sentence and each element within the waveform as a word. Inverting the depth relationships of the elements and the adjacent relationships of the waveforms is essentially a process of “translating” GPR data into permittivity values. Given the complexity of GPR B-scans, as illustrated in Fig. 6, the features extracted by the convolutional neural network are fuzzy, and some of the information is lost. Due to the limitation of the sensory field, the local information is over-amplified or weakened, and it is difficult to establish the global correlation between the waveforms. Compared with the “scarce” compression features captured by the convolutional neural network, the features extracted by the recurrent neural network are clear and obvious, with almost no loss of information. The captured “rich” features fully retain the correlation between anomalous waveforms and neighboring waveforms, and when combined with the contextual information, they can form the continuity of depth direction and global correlation between waveforms. These features can provide more information to establish the mapping between the electromagnetic wave signal and the measured object, thus easily “translating” various materials and diseases in the GPR data into corresponding permittivity.

The entire “wave to permittivity” process is shown in Fig. 7. In GPRTNet, GPR data are first inputted into the encoder for waveform dimension adjustment and data-type conversion. The converted

waveform data are then mapped into a wave vector sequence, called “wave embedding.” Each wave vector represents a time step in the input sequence, with the total length of time steps corresponding to the total number of input radar data traces. The encoding network processes each time step of the wave vector sequence, encoding this information into a vector known as the “wave vector,” which contains the entire waveform sequence information used by the decoder to generate the target model sequence. The wave vectors outputted from the encoder are inputted into the decoder for prediction generation. The decoding network outputs the predicted permittivity value for the current time step and the new network weights by receiving the network weights and the predicted permittivity value from the previous time step. Each time step maps the wave vector sequence to the permittivity sequence. The decoder sequentially decodes the feature wave vectors, completing the “wave to permittivity” process in GPRTNet. Meanwhile, GTN-Conv and GTN-VGG also adhere to the “wave to permittivity” architecture in their design. The encoder extracts features from each GPR waveform and encodes them into wave vectors, which are then translated into permittivity sequences by the GRU decoder.

Notably, in this paper we directly use the raw GPR data rather than relying on B-Scan images for the translation, a choice based on the consideration of fully utilizing the characteristics of the GPR data, aiming at preserving the integrity of the signal and enhancing the accuracy of the translation. The electromagnetic waveform data, as the raw output of GPR, contains real waveform features that maintain the time or depth dimensions, as well as rich information on the original electromagnetic waveform reflection characteristics. In contrast, mapping GPR data into B-Scan images usually requires grayscale processing, a process that inevitably loses some signal details, especially under-representing the weak variations in low reflection intensity regions. The direct use of GPR data avoids these potential problems, allowing the network to learn complete and realistic electromagnetic wave reflection properties and to capture weak feature changes. On this basis, the “Wave to Permittivity” framework takes the electromagnetic wave sequence as the input directly, and the waveform information of each track can be processed independently, avoiding the limitation of data volume or

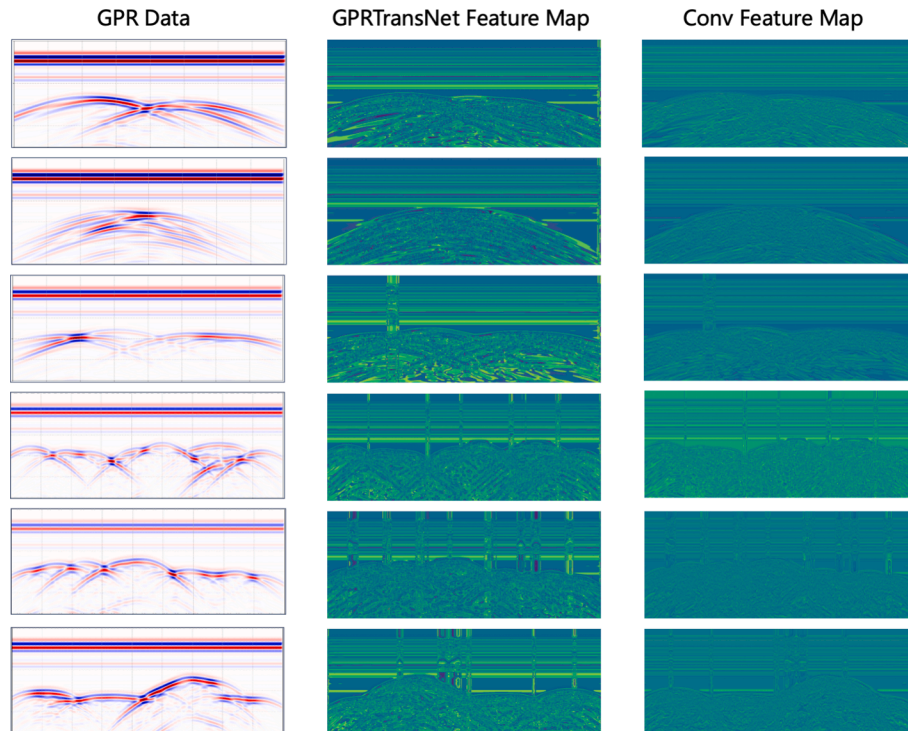


Fig. 6. Comparison of Feature Map.

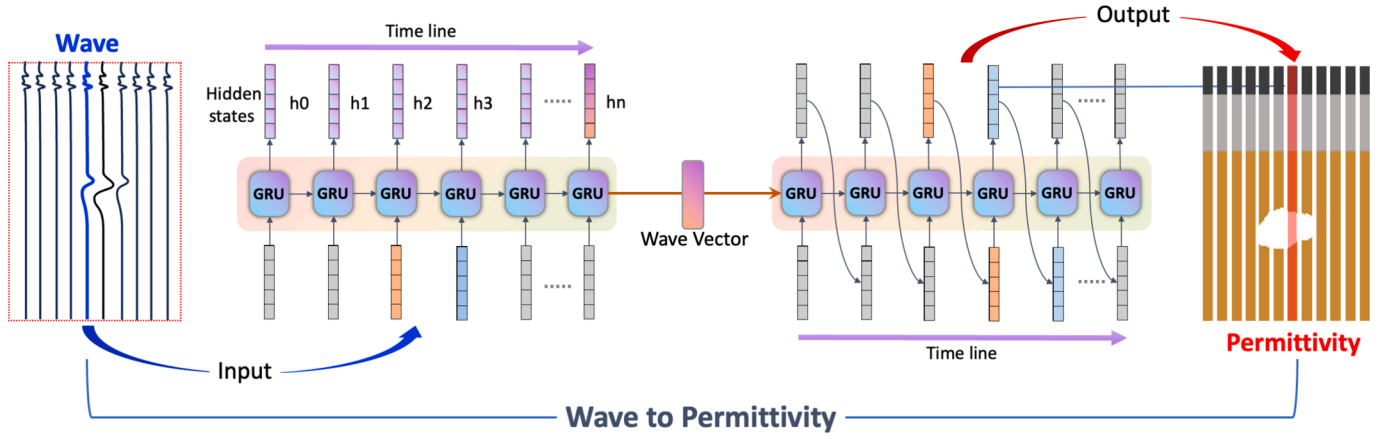


Fig. 7. "Wave to Permittivity" Process.

resolution, and realizing the translation of GPR data of any length, showing strong flexibility and adaptability.

3.3. GPRTransNet with attention

In the previous discussion, GPRTransNet encodes the entire sequence of ground-penetrating radar data into a wave vector, which is a fixed-length vector. The quality of the wave vector directly influences the

decoder's information and ultimately determines the quality of the translation results. Because of the sequential structure of GRUs, the encoder tends to focus more on the elements at the end of the input sequence, causing the wave vector to "forget" the earlier elements of the input sequence. This characteristic allows GPRTransNet to quickly translate radar defect data in simple scenarios but may produce rough results in complex scenarios. To address this, we introduce the attention mechanism to enhance the translation of defects in complex scenarios.

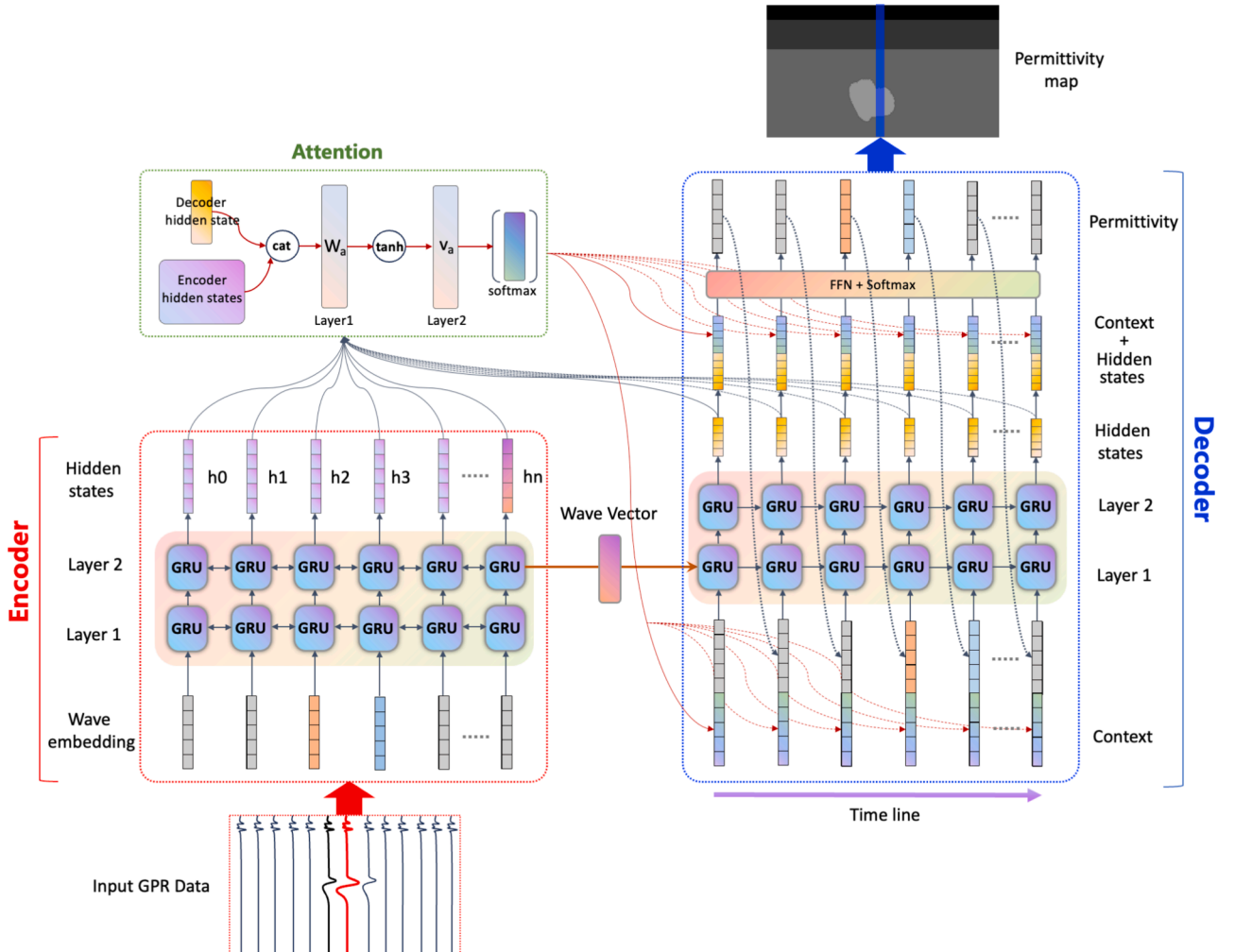


Fig. 8. GPRTransNet With Attention.

As shown in Fig. 8, the enhanced network, called “GPRTransNet-Pro,” increases the network’s complexity with multilayer bidirectional GRU networks of 128 neurons and incorporates the attention mechanism. Attention is a mechanism that enables the model to learn to focus on specific parts of the input sequence during decoding, gathering the semantic details needed to produce the decoder’s output at each time step. Attention weights simulate the brain’s attention allocation mechanism, assigning more attention to more critical parts, thereby increasing focus on defect areas. The calculation process involves using the previous decoder hidden state and all encoder hidden states to perform attention for each decoded permittivity, determining the relevance of the current output to all input elements, which is the weight. A weighted average method then calculates a context vector representation that is more favorable for the current output.

$$\text{score}(\mathbf{h}_t, \bar{\mathbf{h}}_s) = \mathbf{v}_a^\top \tanh(\mathbf{W}_a[\mathbf{h}_t; \bar{\mathbf{h}}_s]) \quad (1)$$

First, the attention score between the previous decoder hidden state and all encoder hidden states is computed, representing the alignment probability of the output permittivity at a specific time with all input radar data. The decoder does not directly take the encoder’s hidden state as an input but uses a selection mechanism to determine which hidden states are most relevant to the current node, combining information from the most relevant hidden states. The attention score is used as a weight to calculate the weighted average of all encoder hidden states, resulting in the context vector for that time. In the decoder, the previous hidden state, the output permittivity sequence, and the current context vector are used as inputs, allowing each decoder node to receive direct information from the encoder rather than only the first node receiving the context vector.

When GPRTransNet-Pro translates current radar data, the network looks for corresponding data in the source waveforms and combines previously translated parts for the translation. For example, when translating a “defect,” it focuses attention on the “anomalous” parts of the source waveform. This approach allows the decoder to see all encoder’s information when predicting permittivity, not limited to the fixed-length hidden vector in the original model, and retains long-range information.

3.4. Encoding and decoding networks

The capability of the RNNs used in the encoder–decoder structure

determines the number of adjacent waveforms from which we can extract information related to anomalies. However, standard RNNs tend to forget information when dealing with long sequences. To address the long-term dependency issues in RNNs, variants such as long short-term memory (LSTM) (Hochreiter et al., 1997) and GRU (Chung et al., 2014) were proposed. In particular, GRU has the advantages of fewer parameters and faster computation speed. As shown in Fig. 9, experimental results indicate that compared with LSTM, GRU provides shapes that better conform to real conditions when translating large defects and performs better in handling smaller defects. Detailed analysis shows that GRU-based encoder–decoder networks produce superior translation results compared with LSTM and GRU is easier to train, significantly improving training efficiency. Therefore, this paper uses GRU as the encoder–decoder network.

The core idea of the GRU network is the introduction of two gating mechanisms: the update and reset gates. As shown in Fig. 10, these gating mechanisms help the network decide how to update the hidden state at each time step, selectively ignoring or forgetting some information from the radar waveform data.

The reset gate controls whether GPRTransNet should ignore previous state information, thereby controlling the flow of information. In GPR data, the characteristics of the reflected signals may only relate to nearby anomalies. The reset gate allows the network to focus more on the characteristics of the current input signal by selectively forgetting the information in earlier hidden states, enhancing the ability to capture short-term dependency features. When dealing with noisy GPR data, the reset gate can effectively filter out irrelevant or noisy information, enabling the network to better learn key features related to permittivity translation. The reset gate is calculated as follows:

$$R_t = \sigma(X_t W_{xr} + H_{t-1} W_{hr} + b_r) \quad (2)$$

where X_t represents the current input, H_{t-1} represents the hidden state at the previous time step, W_{hr} , W_{xr} , and b_r are learnable weight parameters, σ is the sigmoid function, and R_t represents the output of the reset gate.

The update gate controls whether GPRTransNet should retain previous state information and how to combine new input information with previous state information. In GPR translation, the update gate can preserve essential long-term information, ensuring that the network does not forget crucial information when processing long sequences. This is particularly useful for identifying feature changes over long time spans. The update gate can balance maintaining long-term dependencies

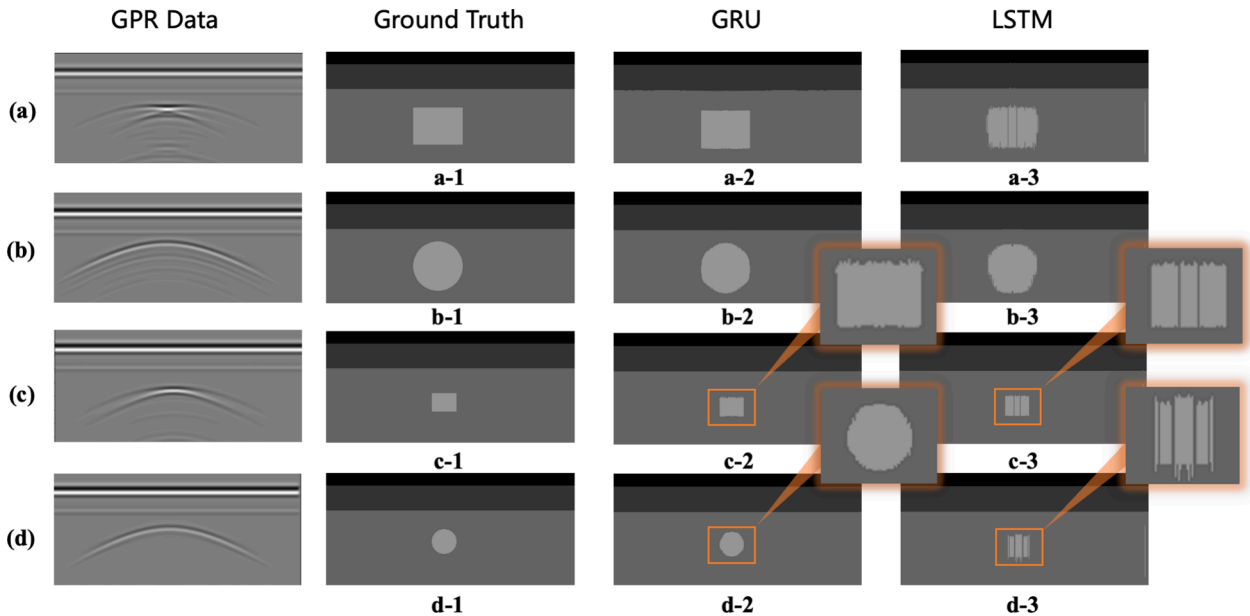


Fig. 9. Comparison of the results of the RNN.

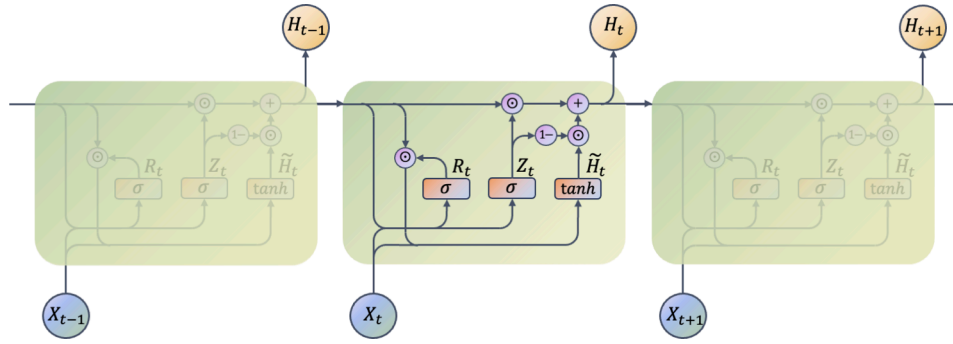


Fig. 10. GRU structure.

and capturing short-term variations, reflecting the overall trend of GPR data while capturing instantaneous local features. The update gate is calculated as follows:

$$Z_t = \sigma(X_t W_{xz} + H_{t-1} W_{hz} + b_z) \quad (3)$$

where X_t represents the current input, H_{t-1} represents the hidden state at the previous time step, W_{hz} , W_{xz} , and b_z are learnable weight parameters, σ is the sigmoid function, and Z_t represents the output of the update gate.

In GRU, the candidate hidden state is computed using the outputs of the update and reset gates. The formula is as follows:

$$\tilde{H}_t = \tanh(X_t W_{xh} + (R_t \odot H_{t-1}) W_{rh} + b_h) \quad (4)$$

where W_{xz} , W_{hz} , and b_h are learnable parameters, $\tanh$ is the activation function, and $\odot$ represents elementwise multiplication.

As shown in Fig. 10, the final update formula for the hidden state H_t is determined by combining the effect of the update gate Z_t on the old hidden state H_{t-1} and the candidate hidden state $\tilde{H}_t$:

$$H_t = Z_t \odot H_{t-1} + (1 - Z_t) \odot \tilde{H}_t \quad (5)$$

In GPRTransNet, the hidden states of the GRU encoder at each time step can capture information from each wave data in the input waveform sequence and between waves, encoding this information into a wave vector. This is crucial for long-span GPR data, ensuring that critical information is not lost during subsequent translation processes. As a decoder network, GRU can receive the hidden state of the previous time step and the permittivity generated in the previous time step at each time step, outputting the predicted permittivity and new hidden state for the current time step, thereby mapping the feature vector to the target sequence. GRU uses the context information provided by the encoder and, through its gating mechanisms, selectively remembers and updates information at each time step, ensuring that the generated permittivity value sequence accurately reflects the input waveform's features. The GRU decoder generates a time-continuous output sequence, ensuring temporal consistency and coherence in the translation results, which is crucial for reflecting continuous changes in tunnel or geological environments.

3.5. Loss function

Mean squared error (MSE) is a commonly used metric in regression analysis to measure the average squared difference between the predicted values and the actual observed values. When evaluating the predictive capability of a regression model, MSE is widely used as a loss function. Assuming that a dataset contains n samples, where y_i is the actual value and $\hat{y}_i$ is the corresponding predicted value, the MSE is calculated as follows:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (6)$$

The cross entropy loss function is another common metric in machine learning used to measure the difference between the model's output probability distribution and the actual label distribution, especially in classification problems. Let n be the number of samples; K , the number of categories; y_{ik} , the actual category label; and $\hat{y}_{ik}$, the model's predicted probability that the i -th sample belongs to the k -th category:

$$\text{CrossEntropyLoss} = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K [y_{ik} \cdot \log(\hat{y}_{ik})] \quad (7)$$

Focal loss is a loss function specifically designed to address class imbalance and the difficulty of classifying challenging samples. Its primary goal is to reduce the impact of easy samples and increase the model's focus on difficult-to-classify samples. Focal loss achieves this by adjusting the weight distribution of samples, lowering the weight of easily classified samples, and increasing the weight of hard-to-classify samples. This helps in improving the model's performance on minority classes or challenging samples. Let n denote the number of samples; K , the number of classes; y_{ik} , the actual class label; $\hat{y}_{ik}$, the model's predicted probability that the i -th sample belongs to the k -th class; and γ , a tunable parameter.

$$\text{FocalLoss} = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K [y_{ik} \cdot (1 - \hat{y}_{ik})^\gamma \cdot \log(\hat{y}_{ik})] \quad (8)$$

3.6. Evaluation function

Perplexity (PPL) is a metric used to evaluate the performance of language models, especially in tasks involving natural language processing and language translation. In the GPR translation task, we use PPL to measure the ability of GPRTransNet to predict the permittivity value given a sequence of GPR data. It quantifies the uncertainty or confusion of the model when making predictions. A lower PPL usually indicates better model fit because it implies that the model can more accurately predict the next permittivity value. PPL is calculated as follows:

$$\text{PPL} = 2^{-\frac{1}{N} \sum_{i=1}^N \log p(w_i | w_1, w_2, \dots, w_{i-1})} \quad (9)$$

where N is the length of the wave sequence and $p(w_i | w_1, w_2, \dots, w_{i-1})$ represents the probability of the model predicting the i -th permittivity value given the distribution of the preceding $i-1$ permittivity values.

Structural Similarity (SSIM) (Zhou et al., 2004) was originally proposed to measure the similarity of two images by comprehensively evaluating the visual coherence between images through luminance, contrast and structural information. In the GPR translation task, the distribution of permittivity values reflects the structural information of the data, but the traditional SSIM evaluation method is mainly based on the image gray value for similarity calculation, which cannot be directly

related to the actual physical meaning and engineering requirements of permittivity. Therefore, in this paper, SSIM is re-targeted and modified, and GPR Structural Similarity (G-SSIM), which is applicable to GPR tasks, is proposed. The core difference between G-SSIM and traditional SSIM is that the former does not focus on the visual similarity between the final output visualization picture and the real model picture, but directly evaluates the structural similarity of the permittivity distribution of the network output. This approach directly measures the degree of match between the translation result and the real permittivity distribution, combines the dual assessment dimensions of numerical error and distribution consistency, and avoids reducing the physical distribution characteristics of permittivity to image features, thus more accurately reflecting the reconstruction ability of the network model for permittivity and the translation performance. G-SSIM comprehensively measures the consistency of different models on permittivity distribution by selecting a local window, calculating the local mean, standard deviation and covariance, and combining this information to derive the similarity index of the overall permittivity distribution. Its calculation formula is as follows:

$$G-SSIM(E_1, E_2) = \frac{(2\mu_{E1}\mu_{E2} + C_1)(2\sigma_{E1E2} + C_2)}{(\mu_{E1}^2 + \mu_{E2}^2 + C_1)(\sigma_{E1}^2 + \sigma_{E2}^2 + C_2)} \quad (10)$$

where

μ_{E1} and μ_{E2} are the local means of the two sets of permittivity values,
 σ_{E1} and σ_{E2} are the local standard deviations of the two sets of permittivity values,
 σ_{E1E2} is the local covariance of the two sets of permittivity values,
 C_1 and C_2 are small constants to avoid division by zero.

4. GPR simulation experiments

4.1. Dataset construction

To fully preserve and use the original characteristics of GPR data, we used raw GPR data without applying gain adjustments or using B-scan images. To provide sufficient data for training GPRTransNet, we performed numerical simulations to generate synthetic data for common types of defects. These defects include voids, cracks and structural honeycombing. The dataset essentially covers the most common types of pavement or tunnel lining defects characterized by irregular shapes. The dataset consists of several different categories of defects, including various large and small regular and irregular defects with asphalt layers, cement layers, and sand layers; defects with steel reinforcement layers; and ultralong irregular defects. The permittivity values are shown in Table 1.

We developed the defect generation algorithm based on GPRMax and Python. We developed automated scripts capable of generating a large and diverse set of simulated data based on input parameters. As shown in Fig. 11, for example, when a large number of tiny voids are generated, the total range size of the disease area can be directly defined by setting parameters to reflect the overall distribution range of the disease. In the total range, the characteristics of small voids diseases are controlled by a series of parameters, including the size of voids, the specific distribution location, and the total number of voids. These parameterized Settings can fully reflect the complexity degree of the disease area in different situations. At the same time, we design a hand-drawn driven generation

mechanism for diseases with irregular shapes. Researchers can draw the outline of the disease directly, and the program will automatically extract the geometric features of the drawn image and convert them into a parametric model of GPRMax. Based on this model, the program further generates variants of multiple sizes and orientations and randomly places them within a set background range. This method not only solves the problem that the shapes generated randomly by the computer do not match the actual situation, but also improves the controllability and authenticity of the simulated data, so that we can generate a large number of disease shapes that are close to the actual situation.

We constructed three simulation scenarios to validate GPRTransNet's translation capability for common defects, minor defects, and long-span models:

Simulation Scenario 1: This scenario had a width of 4.8 m and a depth of 2.2 m, with actual output dimensions of 4.0 m × 1.8 m. The grid size for Simulation Scenario 1 was 0.01 m with a time window of 5e-8 s. Each B-scan contained 400 detection traces, with 2121 time steps per trace. In addition, CMPL was used to eliminate false boundary reflections, occupying 10 grids on each side of the permittivity model. For the background medium, as shown in Fig. 12(a), this paper sets up a multi-layer medium including an asphalt layer, a concrete layer, and a sand and soil layer, and the thickness of each layer is 20 cm, 40 cm, and 120 cm, respectively. In the dataset, we randomly generated various shapes of defects and placed them randomly.

Simulation Scenario 2: This scenario had a width of 0.48 m and a depth of 0.21 m, with actual output dimensions of 0.40 m × 0.17 m. Simulation Scenario 2 was set up with a model containing interference from reinforcing steel and similarly used CMPL to eliminate false boundary reflections. The grid size was 0.002 m with a time window of 5e-9 s. Each B-scan contained 200 detection traces, with 1061 time steps per trace. For the background medium, as shown in Fig. 12(b), two cases are set up in this paper, one is a combination of concrete layer, sand and soil layer, and the thickness of each layer is 5 cm and 12 cm, respectively, and the other is a concrete layer of 17 cm. We generated some simple defect types such as rectangles and circles and placed them randomly.

Simulation Scenario 3: This scenario had a width of 9.6 m and a depth of 2.2 m, with actual output dimensions of 8.8 m × 1.8 m. The grid size of Simulation Scenario 3 was 0.01 m with a time window of 5e-8 s. Each B-scan contained 880 detection traces, with 2121 time steps per trace. CMPL was also used in this scenario to eliminate false boundary reflections. For the background medium, as shown in Fig. 12(c), this paper sets up a multi-layer medium including asphalt layer, concrete layer, and sandy soil layer, and the thickness of each layer is 20 cm, 40 cm, and 120 cm, respectively. We generated various irregular defect shapes and placed them randomly, with defects consisting of voids or cracks.

At the input end of the network, to align with the actual GPR data format, we compressed the original time steps of each trace to 512 time steps in the data preprocessing module. We then divided all data into training, validation, and test sets at a ratio of 8:2:3. In particular, these subsets included 92,800 data samples for training, 23,200 for validation, and 34,800 for testing. Each data sample consists of a GPR waveform sequence as the input and its corresponding permittivity sequence as the output. The dataset is constructed to train the model to map GPR waveforms to permittivity distributions effectively.

4.2. Experimental procedure

Experiments were conducted on a server equipped with an Intel i9-14900KF CPU, an NVIDIA GeForce RTX 4090 GPU, and 128 GB of RAM. GPRTransNet was implemented using the Python programming language. The Python-based scientific computing package PyTorch was used, enabling the utilization of GPUs to accelerate the training process of the deep neural network (DNN) model. Subsequently, through

Table 1
Material properties.

Material	Relative Permittivity	Conductivity (s/m)
bitumen	6	0
concrete	9	0.0001 ~ 0.01
Sand	12	0.00014 ~ 0.05
rebar	300	10 ⁻⁸
defect	1	0

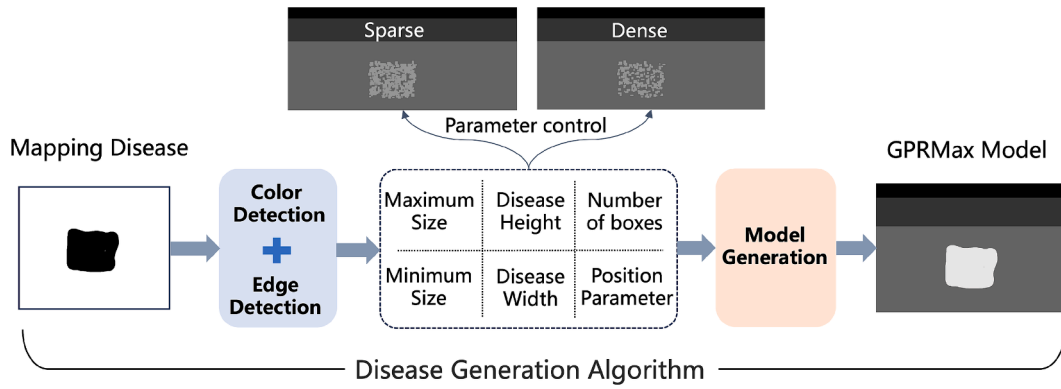


Fig. 11. Disease generation algorithm.

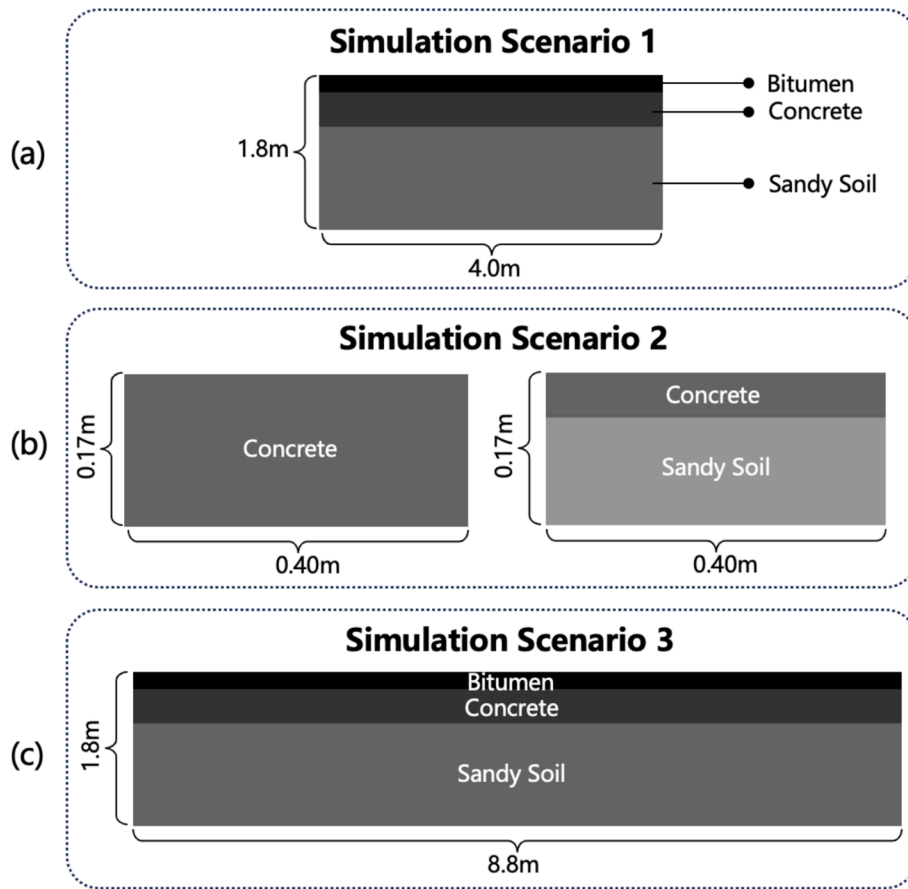


Fig. 12. Composition of background for each type of working condition.

experimental comparisons, this study selected an Adam optimizer with a batch size of 32 and an initial learning rate set to 1×10^{-3} . The DNN model was trained for 500 epochs on the entire training set, indicating that the training process had sufficient epochs to converge. After each epoch, the network's generalization ability was tested using the validation set to prevent overfitting. Finally, the model with the best performance on the validation set was selected as the optimal model and saved for future use.

In this study, multiple evaluation metrics were used to quantify the performance of the network. For loss function selection, MseLoss, CrossEntropyLoss, and FocalLoss were used to quantify the error between the translation results and the ground truth model. PPL was used to measure the model's translation capability, and G-SSIM was used to assess the structural similarity of the translation results. As shown in

Table 2
Experimental results.

Loss	Metrics	GPRTransNet-Lite	GPRTransNet-Pro
MSE	loss	0.00596	0.00312
	PPL	1.00598	1.00312
	G-SSIM	0.98357	0.99260
CrossEntropy	loss	0.02288	0.01307
	PPL	1.02314	1.01316
	G-SSIM	0.98465	0.99248
Focal	loss	0.27028	0.14678
	PPL	1.31013	1.15807
	G-SSIM	0.82380	0.92164

Table 2, when using FocalLoss as the loss function, the loss, PPL, and G-SSIM results for GPRTransNet-Lite and GPRTransNet-Pro were generally mediocre. This could be attributed to the fact that the multiclass inversion task of GPR is inherently very similar or ambiguous and the presence of noise or uneven feature distribution in the data may not align well with the features that focal loss is designed to address, resulting in the poor performance of focal loss.

When MseLoss was used as the loss function, the loss, PPL, and G-SSIM values for GPRTransNet-Pro were 0.00312, 1.00312, and 0.99260, respectively, which were all superior to GPRTransNet-Lite's 0.00596, 1.00598, and 0.98357. When CrossEntropyLoss was used, GPRTransNet-Pro's loss, PPL, and G-SSIM values were 0.01307, 1.01316, and 0.99248, respectively, which also outperformed GPRTransNet-Lite's 0.02288, 1.02314, and 0.98465. The G-SSIM values indicate that the difference in

performance between MSE and CrossEntropy as loss functions is minimal, a conclusion that is corroborated by the translation results.

As shown in Fig. 13, when using FocalLoss, GPRTransNet-Lite performed poorly, and although GPRTransNet-Pro could approximate the shape, it was disproportionately large. By contrast, when using MseLoss or CrossEntropyLoss, the performance was excellent. In particular, GPRTransNet-Pro's translation results outperformed those of GPRTransNet-Lite, with the latter showing varying degrees of artifacts and the Pro version being smoother. The comparative results suggest that regardless of whether MseLoss or CrossEntropyLoss is used as the loss function, GPRTransNet-Lite's translation performance does not vary significantly. However, as shown in Fig. 13 c-3 and Fig. 13 f-3, GPRTransNet-Pro exhibited slightly better results with MSE loss, which is why MSE loss is used in subsequent network experiments.

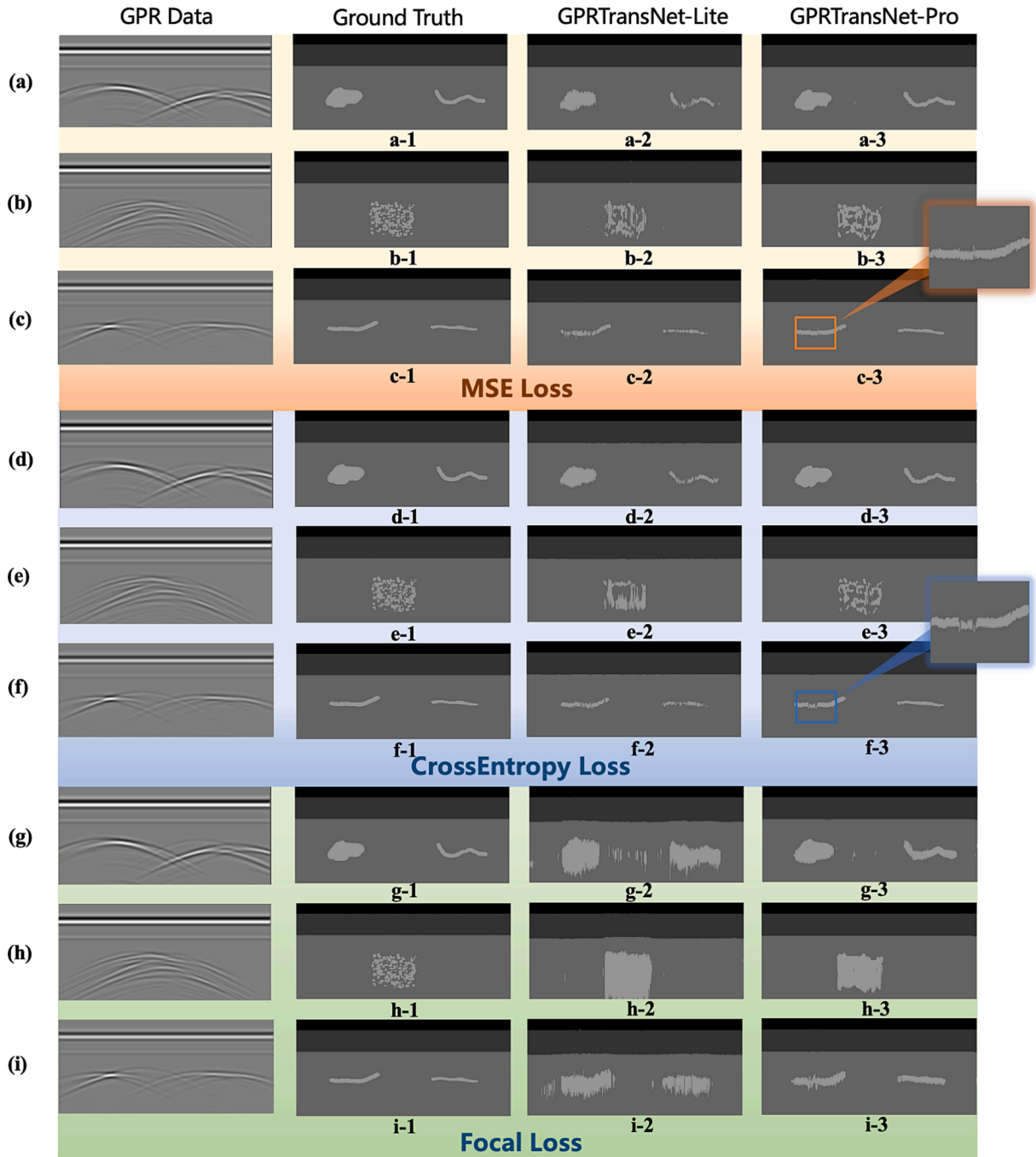


Fig. 13. Comparison of the results of different loss functions.

4.3. Comparison of conventional models

As shown in Fig. 14, this study selected six types of typical defects of varying sizes and shapes to validate the performance of GPRTransNet. Compared to the other models, the Enc-Dec model showed more noise due to its picture-based inversion, and the inverted disease was very fuzzy with no detail. GTN-Conv based on 1-D convolution performed poorly, especially evident in Fig. 14 c-2, where the translated defect shape does not closely match the actual model. GTN-VGG, GPRTransNet-Lite, and GPRTransNet-Pro all accurately translated the shape and position of the permittivity targets and clearly defined the material layers, with the target sizes being almost identical to those of the actual models. However, upon close examination of the details, GPRTransNet-Pro demonstrated superior performance in terms of the smoothness and completeness of defect generation. Nevertheless, because of its lightweight design, GPRTransNet-Lite achieved translation speeds approximately four times faster than those of GPRTransNet-Pro, highlighting GPRTransNet-Lite's advantage in this regard.

As shown in Fig. 15, we further increased model complexity by simulating defects of various types, such as voids, cracks, and non-uniformities. It can be seen that the Enc-Dec model can only invert the approximate location of the performance, and the shape and details of the disease are very poor. GTN-Conv works fine for some of the diseases (Fig. 15 b-3), but most of the cases are no longer quite able to translate the disease information completely. GTN-VGG performs well in most of the results, could provide relatively accurate translations of defect positions and shapes, but in some scenarios, it exhibited issues with missing translation targets (Fig. 15 d-4). Although GPRTransNet-Lite could quickly translate simple defects, it began to exhibit significant detail loss when dealing with more complex situations. For instance, when translating cracks, noticeable artifacts and discontinuities were evident in the translation results. However, GPRTransNet-Lite did not suffer from target omission issues, which was an improvement over the two convolution-based translation networks. GPRTransNet-Pro, leveraging the attention mechanism and the memory capabilities of the bidirectional recurrent network, performed exceptionally well in translating targets. As shown in Fig. 15, the results of GPRTransNet-Pro exhibited rich details, with significant improvements in both the edges and continuity of the defects.

4.4. Permittivity models with steel bars

In practical engineering applications, steel reinforcements, being highly conductive metals, produce strong reflections and scattering when exposed to radar waves. These reflected and scattered signals superimpose on the original echo signals. In addition, electromagnetic waves experience energy attenuation and waveform distortion when encountering metal obstacles, such as steel reinforcements, which can form complex network structures underground. This leads to multiple reflections of electromagnetic waves between reinforcements, resulting in multipath effects. Consequently, the localization and shape analysis of voids become more challenging, affecting the accuracy of void translation and positioning.

To enhance the model's realism, we introduced steel reinforcements to interfere with the defects. We selected cases with only steel reinforcements, defects next to the reinforcements, defects below the reinforcements, and defects below layered reinforcements. The interference of rebar presents a challenge to Enc-Dec, GTN-Conv, and GTN-VGG, and we can see that the loss of detail in Enc-Dec is too severe, with either rebar or disease blurred or missing, and both GTN-Conv and GTN-VGG are able to translate well characterized rebar, but the disease information has been severely lost. GPRTransNet-Lite successfully translates the steel reinforcements and provides a general shape and position of the defects, although with considerable detail loss. As shown in Fig. 16, GPRTransNet-Pro's translation of steel reinforcements and defects adjacent to the reinforcements aligns well with the actual model. However, under complete interference from the reinforcements, there are noticeable fluctuations and artifacts in the generated defects. Despite this, the translation results still reasonably match the position and shape of the defects.

Meanwhile, in order to verify the effect of the size of rebar radius on the disease translation, we set up rebars with diameters of 10 mm, 20 mm and 30 mm to simulate the high-interference scenario, respectively. As shown in Fig. 17, Enc-Dec has a serious loss of detail, the scene with small rebar has the worst inversion effect, and the inverted rebar or disease in the rest of the scenes are also very blurred. The overall performance of GTN-Conv is still not so good, which produces more misclassification of the background, and also translates the rebar poorly, and the shapes and sizes of the disease do not quite match each other. GTN-VGG was able to reproduce the rebar better, and the overall

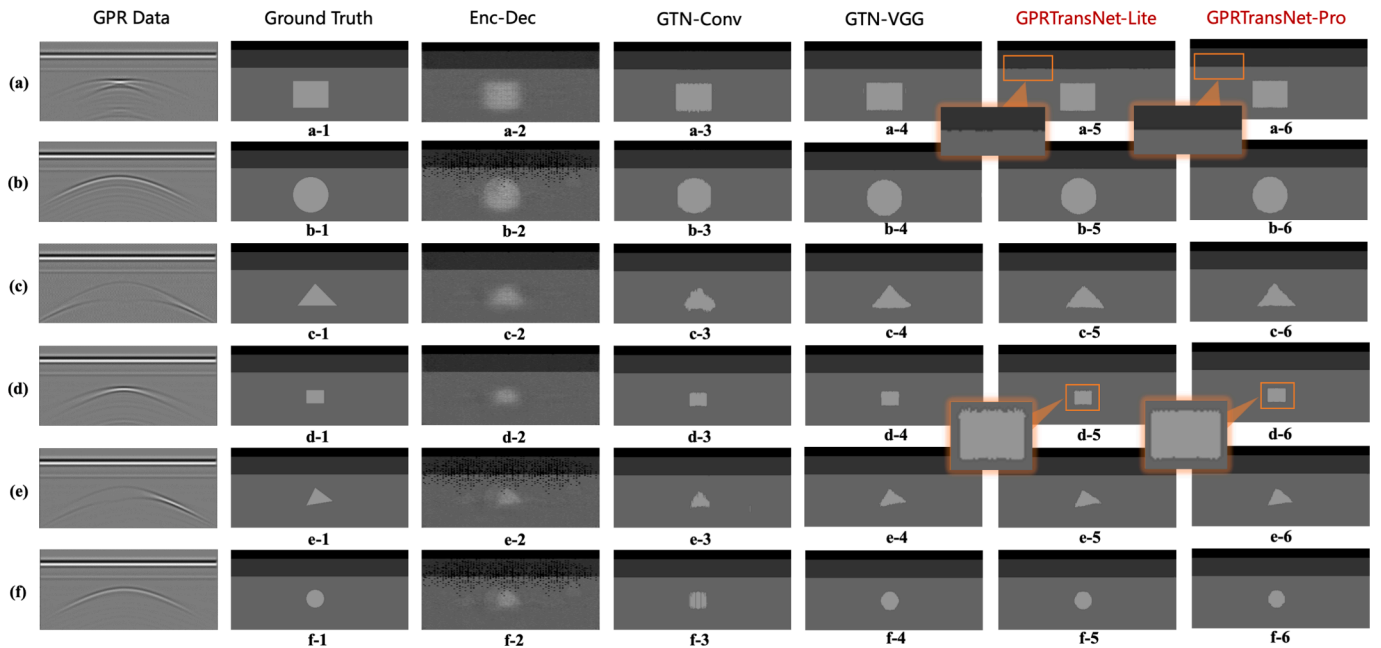


Fig. 14. Comparison of typical geometric defect translation results.

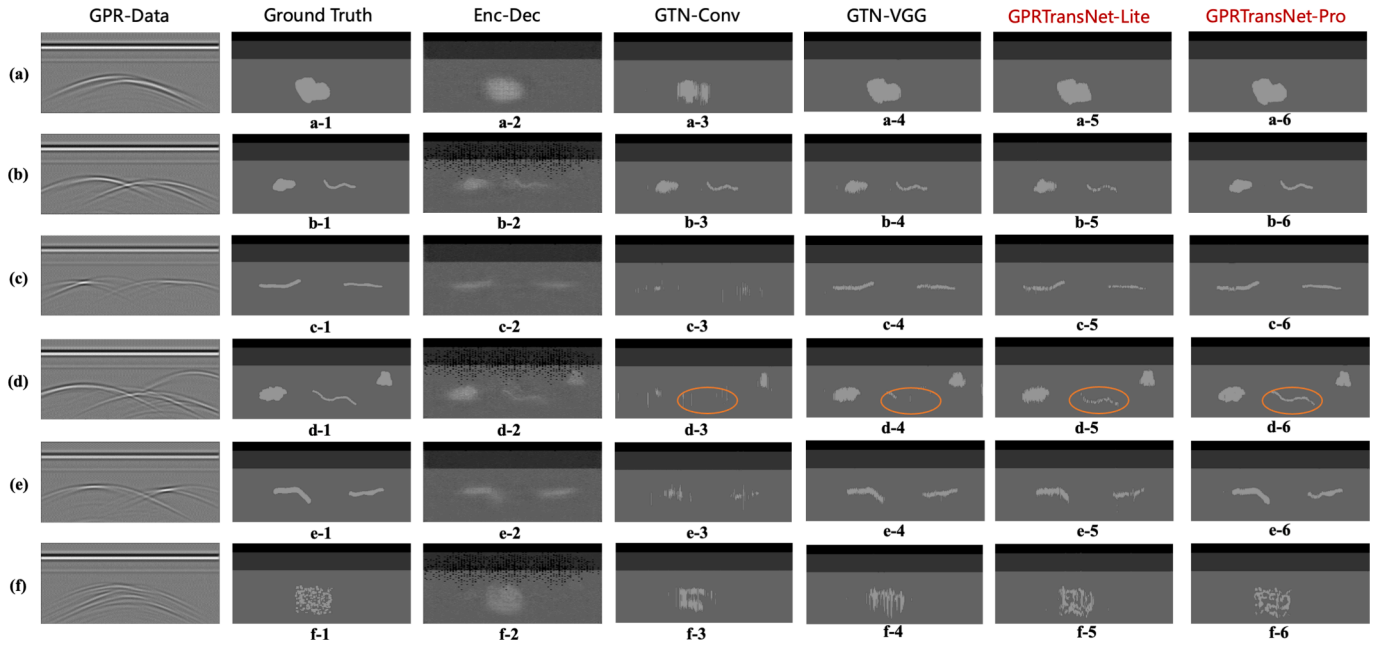


Fig. 15. Comparison of complex defect translation results.

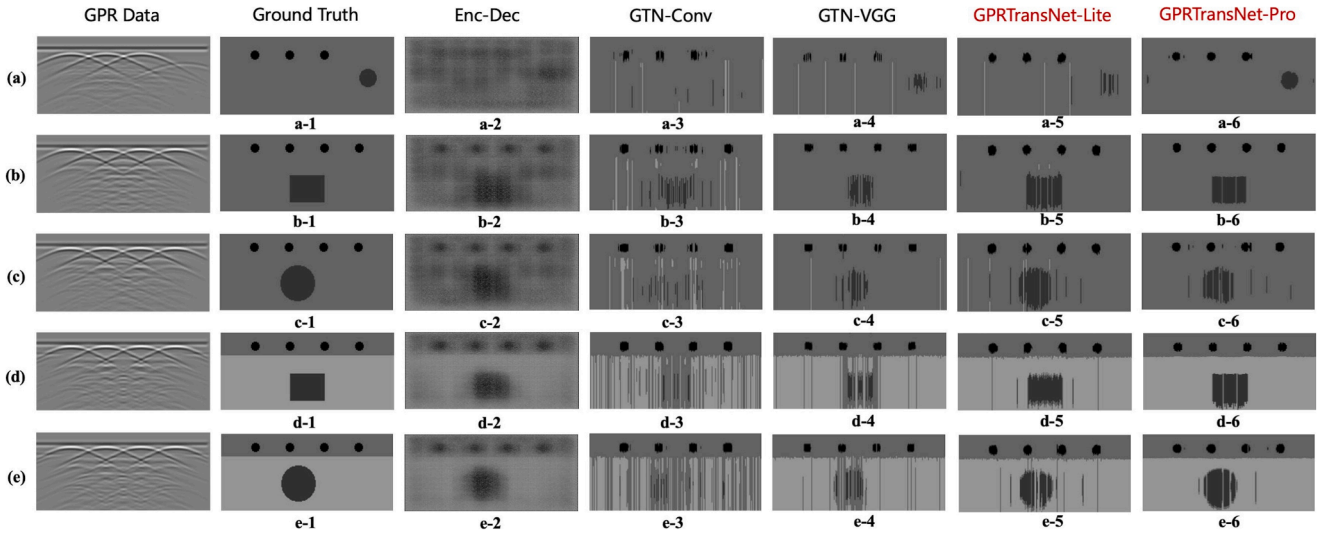


Fig. 16. Comparison of Translation Results with Rebar Interference.

performance was improved but not much. GPRTransNet-Lite translated the rebar, basically unaffected by the size of the rebar, and was able to translate the approximate shape and location of the disease, with a few misclassifications of the background as a drawback. The overall result of GPRTransNet-Pro is the best, which can completely translate the rebars, especially the small-size rebars, and the results of disease translation are not affected by the size of the rebars, and the completeness and size of the diseases are better than the other models, which proves the robustness of the model to different rebar diameters, and comprehensively verifies the performance of the model in the complex scenarios.

4.5. Ultralong permittivity model

To validate the translation capability of GPRTransNet for large models, we also constructed ultralong permittivity models with various types of defects. As shown in Fig. 18, the background material noise of Enc-Dec is obvious, and since the data is segmented inversion, the

diseases appear discontinuous, missing and too fuzzy. The translation results of GTN-Conv suffered from severe detail loss and the defect shapes translated by GTN-VGG are also partly missing. Compared with the two convolution-based deep neural networks, GPRTransNet-Lite and GPRTransNet-Pro exhibited significant “memory” advantages, both demonstrating the ability to translate large models. Although GPRTransNet-Lite generally completed the translation task, it encountered issues with “insufficient memory,” resulting in inaccuracies in defect location, shape, and some missing areas. By contrast, GPRTransNet-Pro maintained a higher level of translation performance. In fact, GPRTransNet-Pro is much more than that. With the advantage of “wave to permittivity”, GPRTransNet-Pro can theoretically translate tens or even hundreds of metres of continuous geological radar data.

The validation results confirmed that GPRTransNet excelled in translating GPR data into permittivity. Furthermore, GPRTransNet-Pro, benefiting from its deeper network layers, bidirectional GRU, and attention mechanism, outperformed GPRTransNet-Lite in translation

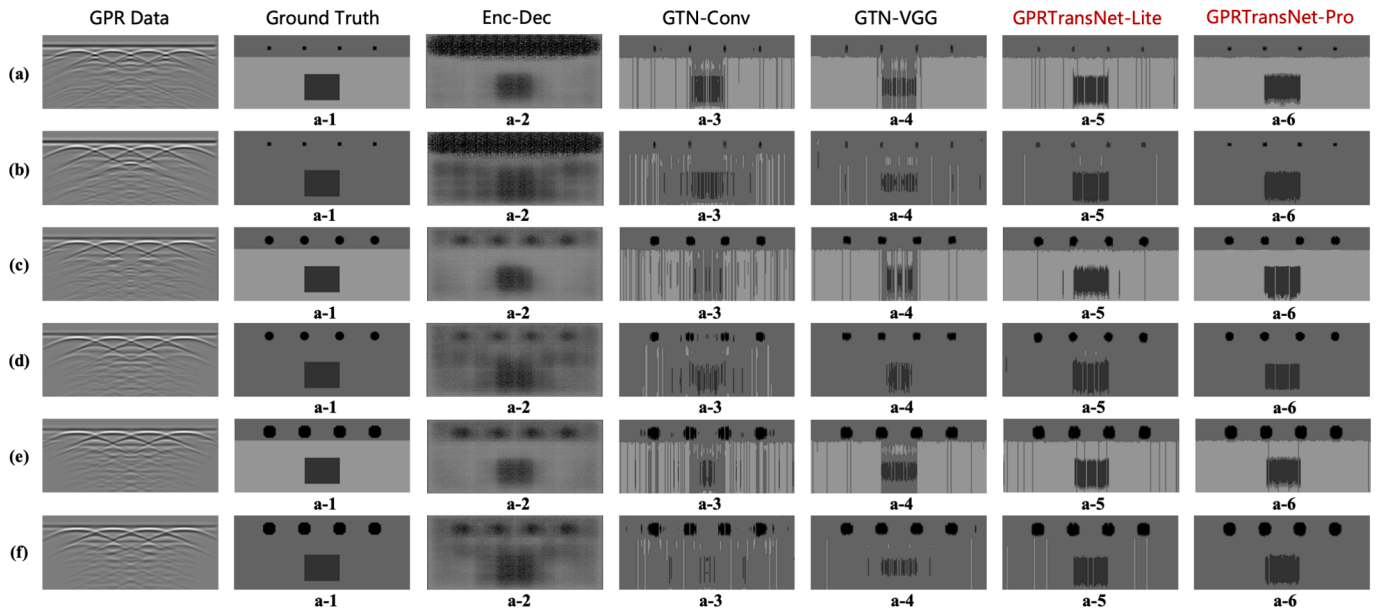


Fig. 17. Comparison of Translation Results for Different Sizes of Rebar Interference.

capability.

4.6. Trace spacing experiments

Trace spacing is an important parameter in GPR data acquisition and refers to the horizontal distance between neighboring swept horizontal distance between neighboring trajectories. Trace spacing directly affects the spatial resolution and amount of data: smaller trace spacing can capture more detailed information, but it also significantly increases the amount of data and processing burden; larger trace spacing may lead to the loss of detailed information, but it can effectively reduce the complexity of data acquisition and storage. In practical applications, different scenes have different demands on trace spacing, so the ability of the model to adapt to changes in trace spacing is especially critical. In order to verify the adaptability of the proposed “Wave to Permittivity” translation framework under different trace spacing conditions, three kinds of trace spacing are set up, namely 0.005 m, 0.01 m and 0.02 m.

As shown in Fig. 19, the most obvious difference between B-Scan and Ground Truth under different trace spacing is the fineness of the data. As can be seen from the translation results, GPRTransNet shows strong adaptability to changes in trace spacing. For a trace spacing of 0.005 m, GPRTransNet fully utilizes the advantages of high-resolution data, and the location, size, and shape of the translated lesions are very fine, especially GPRTransNet-Pro shows high inversion accuracy and boundary recognition ability. When the trace spacing is 0.01 m and 0.02 m, GPRTransNet is also able to reconstruct the defects and material distribution by independently translating each waveform data and integrating the global information, and the results are satisfactory. The experimental results show that with the advantage of the “wave to permittivity” framework of translating electromagnetic wave sequences one by one, the waveform information of each trace can be processed independently regardless of the trace spacing, without the limitation of the overall data volume or resolution, which demonstrates the remarkable flexibility and robustness of GPRTransNet.

5. Real data experiments

In this study, to validate the translation performance of GPRTransNet on real data, we created a 1:1 replica of a tunnel lining concrete model in an experimental field, including structures such as secondary lining, initial support, and steel mesh.

The dimensions of the concrete model were $3 \text{ m} \times 0.99 \text{ m} \times 0.6 \text{ m}$, as shown in Fig. 20. Rebar and cylindrical targets of various sizes, were embedded in the experimental model to evaluate the performance of GPRTransNet. During the experiment, a tape measure and laser distance meter were used to determine the positions of the simulated defects. Excavation was performed at specified locations with a shovel, and PVC cylinders were inserted to simulate void defects, with the bottom end of the cylinder placed at least 10 cm below the radar testing area and the height measured using a laser distance meter. GPR measurements were then conducted with adjusted radar parameters and gain settings, closely adhering to the lining surface, with each experimental condition measured twice.

In this study, the SIR-4000 from GSSI is used as the data acquisition equipment, and the sampling points are set to 512. In order to better adapt to the complex media properties and defect distribution in real scenarios, we introduce a model fine-tuning strategy, which is to optimize the model by inputting part of the dataset into GPRTransNet after processing. Due to the differences between simulated and real data, the direct use of the model trained only on simulated data may lead to performance degradation, so we enhance the generalization ability of the model and reduce the inter-domain gap through fine-tuning to improve the accuracy of the model’s defect identification and permittivity translation under real working conditions. During the fine-tuning process, data enhancements such as random cropping and stretching are performed on the real data in order to enlarge the data size and improve the robustness of the model. A two-phase strategy was adopted for model training, with the initial phase using large-scale simulated data to pre-train the model in order to learn the basic features of electromagnetic wave propagation and defect recognition capability. On this basis, the pre-trained model is fine-tuned using part of the real data so that it can be better adapted to the complexity of the field environment. In the fine-tuning stage, we used a smaller learning rate to ensure a smooth optimization process and stable training results.

The fine-tuned GPRTransNet was then used for inversion on real data, with the inversion results shown in the figures. Fig. 21(a) and (b) depict the inversion results for models with air-filled cylindrical defects of 7.5 and 16 cm in diameter, respectively, positioned at the upper side of the measurement line. Fig. 21(c) and (d) illustrate the inversion results for the same defect sizes but with the measurement line positioned at the middle side.

The translation results revealed that GTN-Conv and GTN-VGG

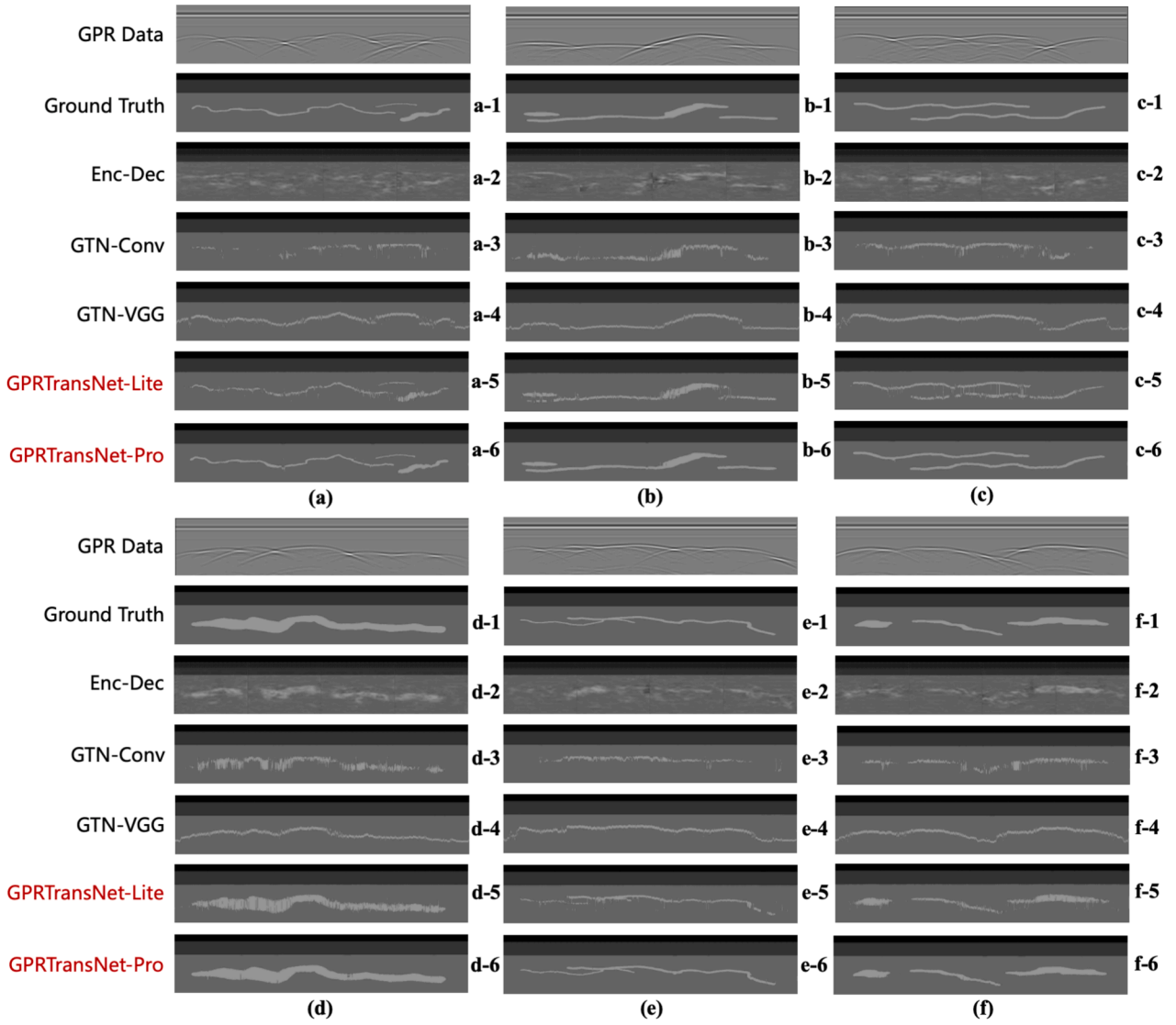


Fig. 18. Comparison of the results of the large model translations.

performed poorly, with both models providing only a very rough representation of the steel reinforcement and defects, with significant missing details. GPRTransNet-Lite provided a general but somewhat vague assessment of the shape and position of defects and steel reinforcements. By contrast, GPRTransNet-Pro demonstrated a strong translation capability, accurately capturing both the size and position of the defects and steel reinforcements. Compared with the steel reinforcement simulation results in the previous section, GPRTransNet-Pro delivered impressive results, with its performance on real data even surpassing that on simulated data.

The primary reason for this discrepancy is the significant interference from steel reinforcement in the simulated data, which made defect features less distinct. Although it was difficult to visually detect the hyperbolic waveform characteristics of the defects from the B-scan images of the simulated data, the real radar data clearly exhibited these characteristics. This validates the feasibility of GPRTransNet for translating real data.

6. Conclusion

This paper presents a novel deep learning network called GPRTransNet that can directly translate raw GPR data into precise permittivity maps. A “wave to permittivity” translation architecture is

also proposed, which is able to translate each wave of GPR into its corresponding permittivity sequences without any limitation of data size, demonstrating remarkable flexibility and robustness. The network is composed of an encoder–decoder structure based on GRUs. The specially designed encoder captures the relationships between anomalous waveforms and adjacent waveforms in GPR data, and the decoder establishes a mapping between electromagnetic wave signals and the object under examination through a condensed “wave vector.” To enhance the model’s ability to express complex environments, we also incorporate an attention mechanism to improve focus on defect areas, enabling the effective “translation” of various materials and defects in GPR data into corresponding permittivity. GPRTransNet shows significant improvement and meaningful advancements in the inversion of GPR.

In addition, this paper introduces a specialized evaluation metric for GPR translation, called G-SSIM, which allows the direct assessment of the translation results. GPRTransNet has been validated on simulated data. The experiments demonstrate that GPRTransNet can clearly translate permittivity maps for various complex defects, including multilayer materials, and can effectively reconstruct basic shapes and positions of defects even under steel reinforcement interference. GPRTransNet has also been applied to real data, with results indicating that it can effectively reconstruct defects with clear boundaries, showing

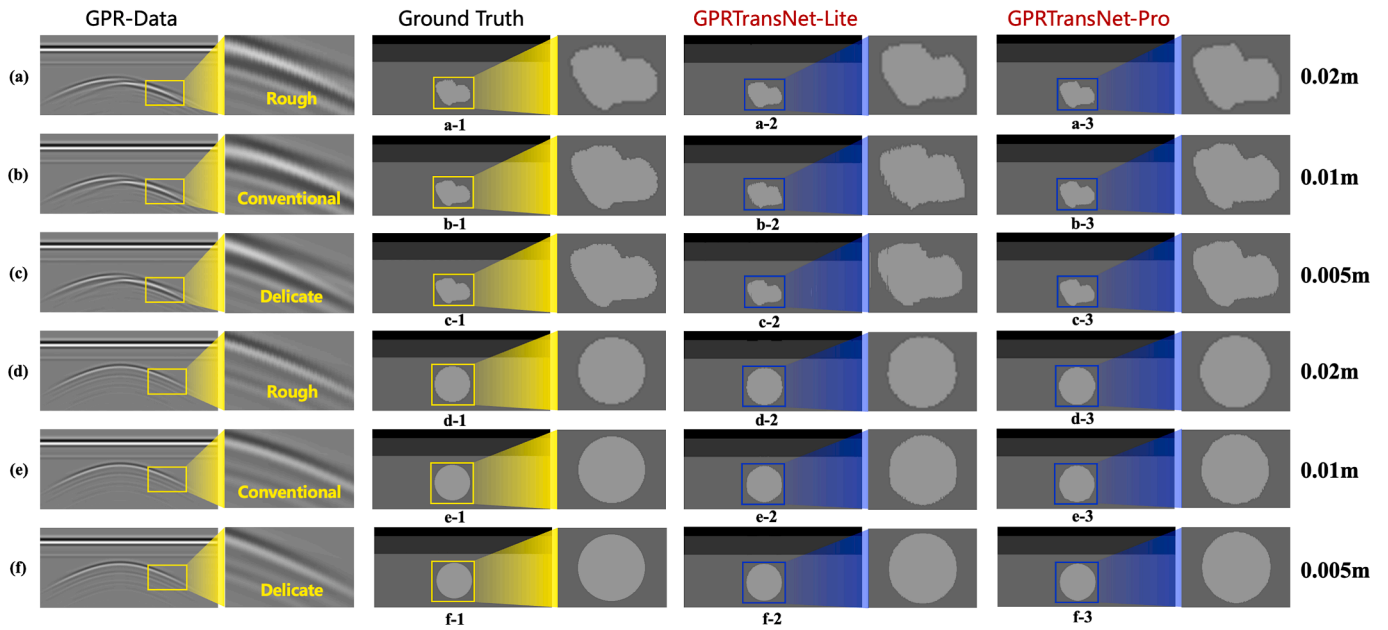


Fig. 19. Comparison of translation results with different trace spacing.



Fig. 20. Field experiment.

satisfactory agreement with real data.

However, because of the strong interference from steel reinforcement models and the complexity of real GPR data, the reconstructed defect contours may be somewhat coarse. In addition, when faced with more densely packed steel reinforcement networks, the network's translation results may become more blurred. Furthermore, because the network relies on previously translated results to predict the next permittivity during the translation process, the inversion speed may be somewhat constrained. These issues warrant further investigation in future research.

CRediT authorship contribution statement

Guanhong Lu: Writing – original draft, Visualization, Methodology. **Lei Kou:** Writing – review & editing. **Pei Niu:** Formal analysis. **Gaohang Lv:** Validation, Data curation. **Xiao Zhang:** Funding acquisition, Data curation. **Jian Liu:** Writing – review & editing. **Quanyi Xie:** Funding acquisition.

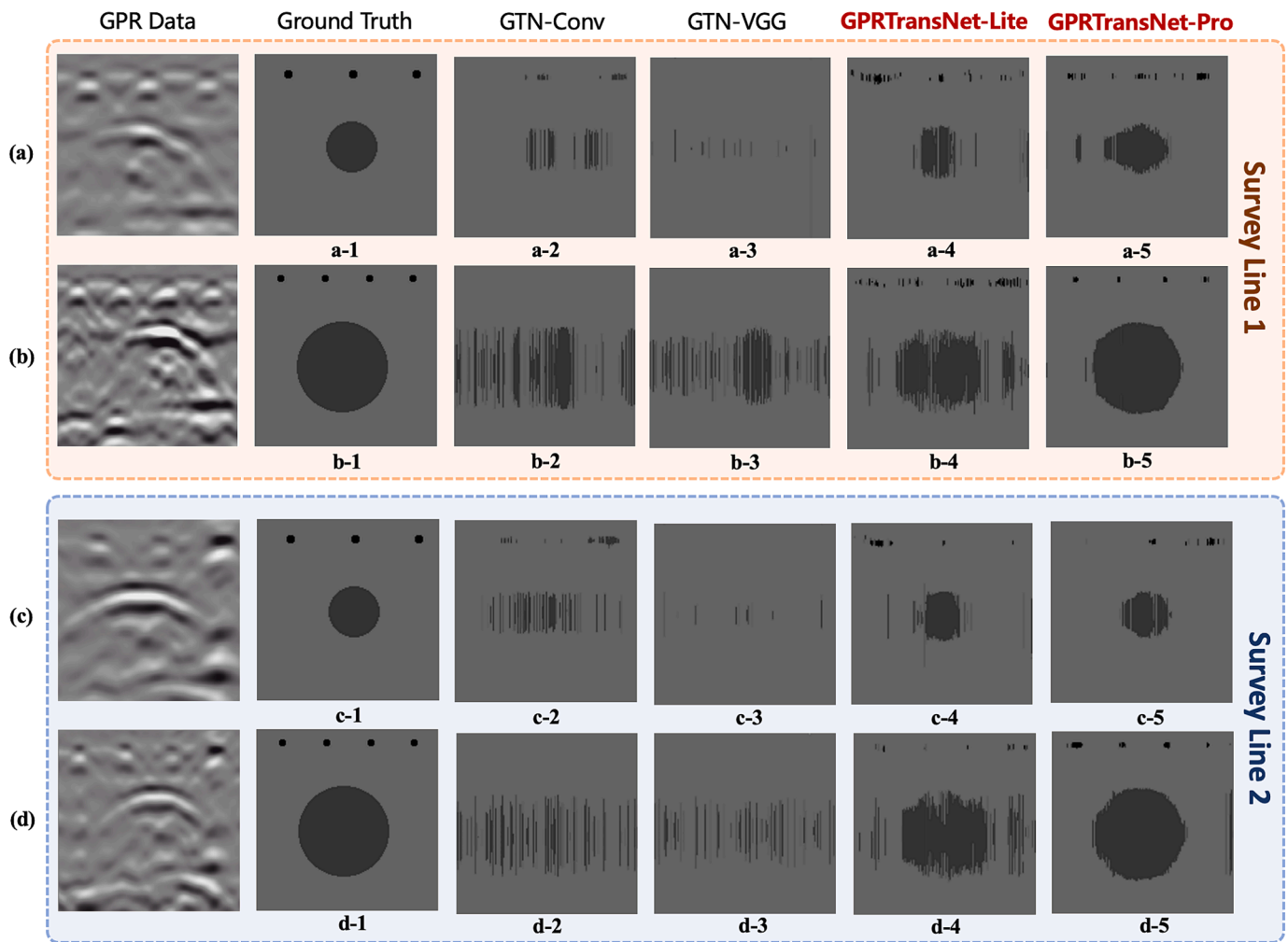


Fig. 21. Result of the real data translation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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